

Sagar Upadhaya

☎ +91-7002643117

✉ sagarupadhaya314@gmail.com



EDUCATION

Assam University, Silchar
M.Sc. in Mathematics

Aug. 2019 - Aug. 2021
CPI: 7.40/10

Cotton University, Guwahati
B.Sc.in Mathematics

Aug. 2015 - Aug. 2018
CPI: 6.33/10

Ramanujan Junior College, Nagaon
Higher Secondary in Science

Jul. 2013 - Jul. 2015
Score: 84.8%

Lanka High School, Nagaon
Secondary Education

Jul. 2013
Score: 82.33%

ACADEMIC PROJECTS

Master Thesis

Jan. 2021 - Jul. 2021

- Title: Riemann zeta function
- Thesis Supervisors: Dr. Naba Kanta Sarma
- In this project, we have studied the Riemann zeta function, its convergence, and the concept of infinite products, which led us to one of Euler's beautiful results, the Euler product. We have also discussed the analytic continuation of the Gamma function and, subsequently, the analytic continuation of the Riemann zeta function using it. In addition, we have proved some related results and explored some recent advances in the study of the Riemann zeta function.

RESEARCH

In this work, I have studied a family of elliptic curves of the form

$$y^2 = x^3 - m^2x + m^2,$$

over the field of rational numbers. The main focus of this research is to investigate rational points and rank-related properties of these curves. Using number theoretic arguments, I have studied double points for certain subfamilies and proved the linear independence of explicitly constructed infinite-order points in $E_m(Q)$ under suitable congruence conditions on m .

PUBLICATION

Sagar Upadhaya, “Rank of the family of Elliptic Curves $y^2 = x^3 - m^2x + m^2$ ”, *Czechoslovak Mathematical Journal* (submitted).

TECHNICAL SKILLS

Sage

Python

Good command of LaTeX

EXTRACURRICULAR ACTIVITIES & ACHIEVEMENTS

- GATE qualified in Mathematics.
- Student Coordinator in the Department of Mathematics, Assam University, Silchar, Sep. 2020 - Aug. 2021.
- Selected from the Department of Mathematics, Assam University, for attending Winter School on Mathematics for Post Graduate Students of the North-Eastern States, held at the Indian Statistical Institute, North-East Centre, Tezpur, during 28th Jan. - 1st Feb., 2020.
- Got 5th rank in the Mathematics Olympiad, level-III, organized by Assam Academy of Mathematics in the year 2013.
- Organized Pi Day in the Department of Mathematics, Assam University, 2020.
- Secured 3rd rank, M.Sc. in Mathematics, 2019-2021 batch, Assam University.
- School Topper in Matriculation board Examination.